



Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure

- ALERT**
- Clozapine requires strict monitoring and precaution in light of its potential to cause neutropenia, agranulocytosis, myocarditis and cardiomyopathy.
 - All patients commenced on clozapine must have a clinical alert entered onto WebPAS. Refer to [Department of Health \(DoH\) WA Clinical Alert MedAlert Policy](#) and [RKPG Clinical Alerts](#).
 - This policy is to be read in conjunction with [Mental Health Charts and Clozapine Resources](#).

Preamble

Consumer Participation

The Rockingham Peel Group (RKPG) recognises and fully supports the need for the patient and/or their primary carer to²:

- Know and exercise their healthcare rights.
- Be engaged in their healthcare.
- Have access to information about treatment options.
- Participate in treatment decisions.
- Have access to information about agreed treatment plans.

Scope of Practice

It is the responsibility of each health care provider to ensure all patient care and therapeutic interventions they provide are within their individual scope of practice. Refer to the [RKPG Guidelines to Writing for Clinical Policy](#) for further information about Scope of Practice.

Purpose

The purpose of this document is to establish safe operational process for patients commencing/recommencing or ceasing clozapine therapy and associated monitoring requirements.

e-Learning

It is recommended that all medical staff involved with prescribing Clozapine complete the e-learning available at [Royal Perth Hospital Clozapine e-Learning](#).

Principles of clozapine therapy

- Treatment with clozapine must be under the supervision of a Consultant Psychiatrist.
- Treatment with clozapine is primarily for patients with a diagnosis of schizophrenia, who are non-responsive to or intolerant to at least two other antipsychotic medications.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			

- The registered use of clozapine in Australia requires that patients are enrolled into a designated monitoring program, which mandates systemic evaluation of haematological parameters.
- Prescribers and pharmacists must be registered with the clozapine monitoring system.
- Clozapine therapy must be started at low doses and titrated slowly to minimise side effects as per suggested titration protocols.

Definitions

ClopineCentral™	The monitoring system is administered by Hospira Pty Ltd. It requires all patients, prescribing doctors, dispensing pharmacists, centre co-ordinators and registered centres using clozapine (Clopine) to be registered with the database. At the time of policy development ClopineCentral™ hold the public hospital tender for clozapine in Western Australia.
ClopineCentral™ Protocol	A set of mandated requirements that must be met for the prescribing and dispensing of clozapine at a registered centre.
Registered Centre	A hospital, clinic or other facility that is involved with the use of clozapine, in conjunction with ClopineCentral™. Patients are listed as 'belonging' to a specific centre through their registration at the centre.
Centre Co-ordinator	Nominated person who is responsible for ensuring the registered centre operates in accordance with the ClopineCentral™ protocol. For Rockingham Peel Group (RkPG), this is the Rockingham General Hospital Senior Pharmacist.

Prior to commencing clozapine

- Patient and carer/support person education.
- Where possible, obtain informed consent from the patient after appropriate discussion of the potential risks and benefits of treatment. Whenever possible, involve the carer or personal support person in this process. Inform and involve any appointed guardian. This information should be recorded within the patient's medical record.
- Provide the patient and carer/support person with relevant patient medication information (CMI) along with the ClopineCentral™ 'Patient Notification Form' as appropriate. [Patient Information Leaflets](#) endorsed by the Department of Health and Office of Chief Psychiatrist are available.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			

Pre-assessment and baseline monitoring requirements

- Blood monitoring to initiate therapy and cease therapy remains the responsibility of the treating Psychiatrist or delegated medical officer (MO).
- Overall, the following baseline investigations are required prior to the initiation of clozapine. Please note that if more than 10 days has elapsed between the Full Blood Picture (FBP) and the initiation of clozapine, another FBP is required as well as another Troponin and CRP.

Table 1 Baseline investigations required prior to commencing clozapine

Full blood count (FBC) including White blood cell (WBC) & Neutrophil count (NC)
Blood group
Troponin
C-Reactive Protein (CRP)
Urea and electrolytes (U+E)
Liver function tests (LFT)
Fasting plasma glucose
Total Cholesterol, HDL, LDL & Triglycerides
12 lead Electrocardiogram (ECG)
Echocardiogram
Pregnancy test where appropriate

Table 2 Baseline measurements prior to commencement

Weight, height and body mass index (BMI)
Waist circumference
Blood pressure (BP) lying and standing, temperature, pulse, respirations

Patient registration with ClopineCentral™

- A pre-treatment FBP - which includes White Blood Cell (WBC) and Neutrophil Count (NC) along with a blood type and a 'Registration Form for New Patients' is completed and submitted to ClopineCentral™.
- Once eligibility for treatment is confirmed through registration with ClopineCentral™, the patient is issued a Clopine Patient Number (CPN). The registration process usually takes up to 24 hours following submission of the 'Registration Form for New Patients'.
- Therapy cannot be commenced until approval of registration occurs.
- Once registered, it is the MO's responsibility to ensure that each patient for whom he/she prescribes clozapine is monitored in accordance with the requirements of the ClopineCentral™ protocol.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Clinical alert for clozapine

- All patients commenced onto clozapine must have a clinical alert completed in their medical record and entered onto WebPAS as per the [DoH WA Clinical Alert MedAlert Policy](#).
- The pharmacist will complete the clinical alert form (MR ALERT 2) and the completed form is scanned and sent to RkPGSystemsSupport@health.wa.gov.au with the completed original form filed in the medical record.
- For patients transferred to the community health service from another health service who do not have a completed clinical alert, it is the MO's responsibility to complete the MR ALERT 2 form and send to RkPGSystemsSupport@health.wa.gov.au with the completed original form filed in the medical record.

Commencement of clozapine

- Patients to be commenced on clozapine can be either be inpatients at Mimidi Park or can be admitted as a 'day patient'.
- For all patients commenced on clozapine, a Clozapine Initiation and Titration Chart (RGMR170R) must be completed.
- The initial use of this chart is restricted to the Mimidi Park Mental Health Inpatient Unit under the supervision of a Consultant Psychiatrist.
- Prior to the first dose of clozapine, the patient will have baseline monitoring observations recorded - temperature, heart rate (HR), blood pressure (BP) (lying and standing), respiratory rate (RR) and level of consciousness. These monitoring observations are to be repeated hourly for the first six (6) hours.
- For inpatients, these observations are continued every six (6) hours for the first twenty four (24) hours.
- Monitoring observations are to be completed twice daily for the first seven (7) days, then daily for the next twenty one (21) days of treatment for a total of twenty eight (28) days of monitoring.
- These observations are to be documented on the Clozapine Initiation and Titration Chart (RGMR170R).
- For 'day patients' see monitoring section below.

Initiation of clozapine as a 'day patient' in Mimidi Park

- A patient can be admitted as a 'day patient' to commence clozapine therapy.
- All relevant baseline blood tests, investigations (see Table 1) must be completed and the patient must be registered on ClopineCentral™ with a Clopine Patient Number (CPN) before being admitted to the ward as a 'day patient'. It is the treating Psychiatrist's responsibility to ensure this occurs.
- The consent and registration and pre-assessment requirements are the same as an 'inpatient'.

How to arrange a bed for 'day patient' initiation

- The Consultant Psychiatrist / Medical Officer / Registrar or Case Manager must liaise with the ECT Coordinator, Mental Health ward pharmacist (Page 374) and Nurse Unit

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Manager (NUM) (extension 4943), including by email, to alert them to a pending day patient initiation for clozapine.

- The ECT Coordinator will liaise with the NUM to determine when a day patient initiation slot will be available considering the current demand on the in-patient unit. (Wednesday is the preferred day with Mondays and Fridays as back up days).
- Each patient must have an allocated community Case Manager who is responsible for providing all clozapine pre-assessment and baseline monitoring results including bloodwork and copies of baseline ECG and Echocardiogram.
- The community case manager is responsible for arranging clozapine follow up appointments for the patient including arranging weekly follow up appointments for the patient in the community.
- If the patient is not commenced on clozapine within **10 days** of the pre-treatment baseline investigations being completed, a FBP, Troponin and CRP will need to be repeated. The results of the FBP will be submitted to ClopineCentral™ and the patient reapproved before being admitted to the ward.
- The community MO or Case Manager is responsible for emailing relevant information and results, including most recent copy of FBP (completed within 10 days of commencement), ECG, echocardiogram, signed clozapine consent form, copy of Notice of Patient registration with ClopineCentral™, Mental Health Assessment form, Risk assessment form to the following email address:
 - * PaRKMHSInpatientUnitTriage@health.wa.gov.au
- The Clozapine Initiation and Titration Chart (RGMR170R) is completed by the inpatient Medical Officer / Registrar and the form is signed by the inpatient Consultant whom the patient is being admitted under as a day patient.

The monitoring for a 'day patient' is the same as for an 'inpatient for the first 6 hours

- After six (6) hours of clozapine monitoring has been completed and the results reviewed as being acceptable the patient will be discharged home with a seven (7) day supply of clozapine issued from Rockingham General Hospital (RGH) Pharmacy. The prescription should be written the morning of admission to allow sufficient time for processing and preparation of the prescription for discharge.
- If the results are not acceptable, the patient should be medically reviewed to determine if it is necessary to remain as an 'inpatient' until their clinical condition has stabilised.
- The Clozapine Initiation and Titration Chart (RGMR170) is emailed to the appropriate community teams for Rockingham Kwinana Adult, Peel Adult and PaRK Older Adult as follows:
 - * PaRKMHSRKDischargeFollowUp@health.wa.gov.au
 - * PaRKMHSPeelDischargeFollowUp@health.wa.gov.au
 - * PaRKMHSOAMHSTriage@health.wa.gov.au
- The patient will be seen for the next 28 days for daily observations, on working week days, of temperature, HR, BP and RR. Please note that a referral to the Intensive Day Therapy Unit (IDTU) or the Mental Health Hospital in the Home (HITH) may be considered as an option for post initiation follow-up. Please note that observations on weekends and public holidays are optional and based on patient need and

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

course. If observations are required on weekends and public holidays these can be undertaken by the Assessment and Treatment Team (ATT) or HITH.

- Observations from day 1 to 7 are to be entered onto Clozapine Initiation and Titration chart (RGMR170). Within the community observations from day 8 – 28 are entered onto the Clozapine Initiation Day 8 – 28 O/P Observation Chart (RGMR170S).
- After hours, ATT/HITH will contact the on-call duty MO or on call Psychiatrist and discuss if any observations fall outside of normal parameters.

Ongoing patient monitoring

- All patients receiving clozapine will have had baseline investigations and measurements recorded and documented.
- For patients initiated on clozapine (or for re-titration where clozapine therapy has been ceased for greater than 4 weeks), a weekly FBP is required for 18 weeks, then every 28 days (monthly) thereafter.

Ongoing investigations following commencement of clozapine are as follows:

- Bloods and investigations are also completed when clinically indicated.
- Weight, BP (lying and standing), body mass index (BMI), waist circumference, HR and temperature (T) should be measured at each visit.
- Ongoing monitoring must be completed on the appropriate form(s).
 - * Mental Health Monitoring Tool for Metabolic Syndrome (RGMR142).
 - * Mental Health Clozapine Monitoring Protocol (RGMR142A).

Table 3 Frequency and tests required

Frequency	Tests required				
Week 1	FBP	Troponin	CRP	ECG	Weight, BP, BMI, waist circumference, HR, T
Week 2	FBP	Troponin	CRP	ECG	Weight, BP, BMI, waist circumference, HR, T
Week 3	FBP	Troponin	CRP	ECG	Weight, BP, BMI, waist circumference, HR, T
Week 4	FBP	Troponin	CRP	ECG	Weight, BP, BMI, waist circumference, HR, T Serum clozapine level
Week 5	FBP	Troponin	CRP	ECG	Weight, BP, BMI, waist circumference, HR, T
Week 6	FBP	Troponin	CRP	ECG	Weight, BP, BMI, waist circumference, HR, T
Week 7-18	FBP weekly				Weight, BP, BMI, waist circumference, HR, T Serum clozapine level as indicated
Week 19+ Every month	FBP	Glasgow Antipsychotic Side Effect Scale – For			Weight, BP, BMI, waist circumference, HR, T Serum clozapine level as

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			

		Clozapine (GASS-C)	indicated
6 months + every six months thereafter	Troponin CRP U+E LFT	Fasting glucose, Total cholesterol, HDL, LDL, Triglycerides	Serum clozapine level
6 months + annually thereafter	ECG		
1 year	Echocardiogram		
2 years	Echocardiogram		
5 years and every 5 years thereafter	Echocardiogram		
Side effects and smoking status must be checked for at every clinical review			

- Once the patient is initiated on clozapine they must be assigned a Case Manager and receive regular follow-up and appointments.
- Patients have their blood work completed on a weekly or monthly basis at a convenient local laboratory close to them.
- For patients who require it, a home visiting blood service can be arranged via the local laboratory.

Normal blood results

- The patient is given a one week, or up to a maximum one month, supply of clozapine.
- Prescriptions are written on PBS Authority prescription forms; the appropriate streamlined code is **5015** for each single supply of clozapine dispensed through the hospital or select community Pharmacies.
- Once the patient has completed 18 weeks of weekly blood tests and is deemed to be suitable for community Pharmacy dispensation of clozapine a prescription can be written with the streamline code **4998**. This enables the patient to obtain a clozapine supply from any willing community Pharmacy.
- The treating MO must complete a Clozapine Blood Count Record Form documenting the WBC and neutrophil count along with a prescription written for a supply of clozapine within 48 hours of the bloodwork being completed.
- The completed Clozapine Blood Count Record Form is taken by the patient with the prescription for clozapine.
- The prescription is dispensed at the RGH or local community Pharmacy.
- The pharmacist also checks blood results, and if normal dispenses the clozapine prescription.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

- The pharmacist signs the Clozapine Blood Count Record Form and forwards to the Clozapine Coordinator who enters completed blood count results into ClopineCentral™ database within 48 hours of blood work being completed.

Monitoring for agranulocytosis and neutropenia

- White Cell Count (WCC) & neutrophil count must be monitored weekly for the first 18 weeks of treatment and 4-weekly thereafter for the duration of treatment.
- More frequent monitoring is required as per the monitoring protocol if WCC and/or neutrophil count declines or if the patient exhibits clinical symptoms of neutropenia or agranulocytosis (e.g. fever, sore throat).

Table 4: Recommended Actions for Blood Results Registered in
Clopine Blood Results Monitoring System (ClopineCentral™)

Clozapine Blood Results Monitoring System		Recommended Action	
		Prior to Initiation	Ongoing monitoring
Green Range	WBC greater than $3.5 \times 10^9/L$ and Neutrophils greater than $2.0 \times 10^9/L$	Clozapine therapy maybe commenced subject to assessment by the treating medical officer and successful registration with a Clozapine Patient Monitoring System (e.g. ClopineCentral™).	Continue clozapine therapy.
Amber Range	WBC $3.0 - 3.5 \times 10^9/L$ And/or Neutrophils $1.5 - 2.0 \times 10^9/L$	Repeat blood count after one (1) week. If still within same range, clozapine therapy may commence subject to assessment by the treating MO and successful registration.	Continue clozapine therapy with twice-weekly blood tests until return to 'green' range.
Red Range	WBC less than $3.0 \times 10^9/L$ And/or Neutrophils less than $1.5 \times 10^9/L$	DO NOT START THERAPY Seek haematologist advice	STOP CLOZAPINE THERAPY IMMEDIATELY Contact haematologist and Clozapine Patient Monitoring Centre (ClopineCentral™).

Amber blood results

- Initiate twice weekly blood tests until WBC and/or neutrophils return to 'green' range. Encourage appropriate physical activity before blood test (demargination of neutrophils).
- Liaise with the Mental Health ward pharmacist to determine other medicines that may be implicated.
- Monitoring requirements following discontinuation of therapy and interruption of therapy are outlined in the respective sections below.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			

Red blood results

- **Stop clozapine immediately and contact the Haematologist and ClopineCentralTM**

Monitoring for myocarditis

- Troponin I and CRP assay must be repeated on days 7, 14, 21 and 28. The ECG and medical assessment must be repeated at day 7 and 14 and then as clinically indicated. The patient must be monitored for clinical features of myocarditis/infective illness throughout treatment, but particular vigilance should be paid during the first four weeks of treatment.
- If clinical features of myocarditis/infective illness (e.g. fever, tachycardia, dyspnoea, chest pain) develop, the Troponin I, CRP and ECG must be repeated daily until the condition resolves or an alternative cause is identified.
- If CRP 50-100mg/L or there is a mild troponin elevation (≤ 2 ULN) clozapine prescribing can be continued with daily troponin, CRP and ECG until the abnormality resolves. Do not increase the clozapine dose during this time.
- If CRP >100mg/L or Troponin > 2X ULN clozapine must be ceased and an echocardiogram and assessment by a physician or cardiologist must be requested.

Monitoring for clozapine Induced gastrointestinal hypomobility (CIGH)

- Bowel habits will be monitored weekly as a minimum requirement for the first 18 weeks of clozapine treatment and then four weekly thereafter, for the duration of treatment. All hospital inpatients require stool charts to be used and patients need to be asked about their bowel habits using the Rome III criteria. Prophylactic laxatives should be considered for all patients prescribed clozapine and extra vigilance is required when other medications that cause constipation are added.
- Monitoring should be based on the following 'Rome III assessment criteria for functional constipation' and must include at least two of the following:
 - * Straining for at least 25% of defecations
 - * Lumpy or hard stools for at least 25% of defecations
 - * Sensation of incomplete evacuation for at least 25% of defecations
 - * Sensation of anorectal obstruction/blockage for at least 25% of defecations
 - * Manual manoeuvres to facilitate defecation for at least 25% of defecations (eg digital evacuation, support of pelvic floor)
 - * Fewer than three defecations per week
- Assertive treatment of any emerging constipation is necessary including a review of other medication that may be contributing. Signs and symptoms that warrant immediate medical attention include moderate to severe.

Therapeutic dose range of clozapine

- Usual dosage is 200-450mg/day.
- Recommended maximum daily dose is 600mg.
- Dosing up to but not exceeding 900mg/day has been used for exceptional cases only.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

- Doses > 450mg increase the possibility of side effects and adverse events including seizures. Careful plasma monitoring along with serum clozapine levels should occur.

Dose titration, maximum dose, drug interactions and side effects

- A suggested dose titration schedule is available on the inside of the Clozapine Initiation and Titration Chart (RGMR170R).
- Patients greater than 65 years of age may require a slower dose increase titration regimen.
- For titration beyond 200mg/day:
 - * If well tolerated, the daily dose may be increased slowly in increments of 25-50mg (maximum 100mg/week).
 - * Many adverse effects of clozapine are dose dependent and associated with a faster titration schedule.
- In most patients, antipsychotic efficacy can be expected with 200 to 450mg/day in divided doses. If the daily dose does not exceed 200mg, a single administration in the evening may be appropriate, especially for those experiencing day time sedation side effects.
- It is important that a therapeutic plasma level is obtained in most patients to maximise the likelihood of therapeutic benefit. The average dose at which optimal plasma levels are reached varies according to gender and hepatic metabolic pathways which are greatly affected by smoking status.
- It is worth noting that a potential cause of non-response to clozapine is through inadequate plasma concentrations.
 - * Assessing plasma concentration can be also very helpful when determining a patient's adherence to treatment.
- The final dose of clozapine should not be solely determined by plasma concentration. However, when a patient appears to not be responding adequately to treatment it is worth ensuring that the achieved plasma concentration is at least 350 microgram/L, with a range of up to 1000 microgram/L as tolerated. In these patients it is also important to determine if any pharmacotherapeutic issues exist (e.g. smoking: excess caffeine intake: poor adherence: drug interactions etc) that may affect the ability to achieve a therapeutic level and to address these factors to the greatest extent possible.
- Clozapine is associated with a number of potential side effects as listed including the small risk of blood and cardiac abnormalities. See Appendix A.
 - * Please note **constipation** should be considered a serious adverse effect, as it can have potentially fatal complications.
- Clozapine is metabolised by CYP1A2; administration with drugs that inhibit this enzyme may increase its concentration and risk of adverse effects, while administration with drugs that induce it may decrease its concentration and possibly its efficacy.
- Of note is that some antidepressants may increase clozapine levels. This is most effectively through the inhibition of the P450 CYP1A2 pathway which is reliably inhibited by fluvoxamine leading to an increase in serum clozapine levels. Fluoxetine inhibits CYP2D6 and increases serum clozapine levels less reliably.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

- Although these interactions should be approached with caution, it is also possible to prescribe these antidepressants with clozapine to cautiously increase serum plasma levels in patients who are not responding to clozapine and have sub-therapeutic plasma levels (i.e. below 350 microgram/L). This undertaking may lead to a clinical improvement and should occur with a gradual increase in antidepressant dose and careful monitoring of serum levels.
- The addition of fluvoxamine can lead to an unpredictable increase in serum clozapine levels. The dose of clozapine to which fluvoxamine can be added should be no more than 200mg. Thus, a dose of fluvoxamine 25 – 50mg per day can be added to clozapine 200mg daily to achieve a therapeutic clozapine serum level. When fluvoxamine is added to clozapine there should be careful monitoring of clozapine related side effects and weekly serum clozapine levels performed until the doses of both clozapine and fluvoxamine are stable. It should be noted that generally the combined dose of clozapine and fluvoxamine need not exceed 300 – 350mg daily to achieve a serum clozapine level within a therapeutic range of 350 microgram/L to 1000 microgram/L.
- Tobacco smoking status should be ascertained at the commencement of therapy as this will increase clearance of clozapine; the dose may need to be adjusted if a patient starts or stops smoking (by up to 50% in some cases).
- Concomitant use of other medications with the potential to cause agranulocytosis is discouraged and increased vigilance should be maintained where this is necessary. Carbamazepine should never be added to treatment with clozapine as both of these medications may cause agranulocytosis. Any current oral antipsychotic medication should be reviewed and reduced or ceased where possible before commencing clozapine.
- When switching from a long-acting antipsychotic injection it is recommended that clozapine should be commenced, in place of the injection, on the date that the injection was next due.
- Concomitant use of long acting depot antipsychotics is not contraindicated but ideally should be avoided.

Interruption of therapy

- ClopineCentral™ must be notified of any therapy interruption, by phone or completion of a 'Therapy Event Form'.
- Haematological monitoring requirements will change depending on the patient's current monitoring frequency. All patients recommencing clozapine following an interruption in treatment greater than 48 hours must have potential dosing alterations and ongoing monitoring as per Table 5.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			

Dosing recommendations if clozapine dose is missed for greater than 48 hours

Table 5 Blood test monitoring schedule following interruption of therapy

Period of interruption (time since last dose taken)	Dosage	Monitoring requirements
Less than or equal to 48 hours	No change to dosage	No change to monitoring
Greater than 48 hours and less than or equal to 72 hours	Start on 12.5mg once or twice a day. If this is tolerated, it may be feasible to titrate the dose to the therapeutic level more quickly than is recommended for initial treatment	No additional monitoring requirements
Greater than 72 hours and less than or equal to 28 days	Start on 12.5mg once daily and rapidly titrate up as per new patient	For weekly monitored patients: Continue weekly monitoring for 6 weeks or as long as needed to ensure a total of 18 weeks: whichever is the greatest. For four weekly monitored patients; Continue weekly monitoring for six weeks. If no abnormality, resume four-weekly monitoring
Greater than 28 days	Restart patient with a New Patient Registration Form. Start at 12.5mg once daily and titrate up as per new patient.	Commence FBP monitoring as per a new patient.

- It is important that daily compliance is monitored when prescribed clozapine. If clozapine is missed for > 48 hrs a psychiatric review is required prior to recommencing therapy.
- Current guidelines recommend recommencing therapy at 12.5mg once or twice daily on the first day however re-titration of dose to therapeutic level depends on each individual and it may be feasible to increase the dose more rapidly under supervision of the treating Psychiatrist.

Clozapine re-challenge

- If clozapine is ceased as a result of neutropenia or agranulocytosis, rechallenge requires haematologist approval as per the monitoring protocol.
- Where patients have been trialled on clozapine and ceased for non-haematological reasons rechallenge is at the discretion of the treating psychiatrist. Reasons for clozapine cessation and why a retrial is considered appropriate along with appropriate consultation, such as cardiology for previous myocarditis, must be documented in the patient's medical record.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			

Transfer of a patient to another facility

- When a patient treated with clozapine at Peel and Rockingham Kwinana Mental Health Service is discharged or transferred to another facility, the patient must be transferred to the registered centre taking over their care by submitting a 'Patient transfer form' either via fax or the ClopineCentral™ website.
- If handover of additional information to the receiving site is required, this can be communicated by adding a note to the patient profile on the ClopineCentral™ website. The centre co-ordinator at the receiving site will then be informed of the transfer via email.

Discontinuation of therapy

- The decision to cease therapy must be made in consultation with the treating Mental Health team wherever possible:
 - * Relapse following abrupt cessation of clozapine can be refractory to treatment.
 - * Side effects to clozapine are not necessarily a contraindication to continued treatment.
- Monitoring of haematological parameters is required after cessation of clozapine.
- Frequency and duration depends on monitoring requirements in place at the time of discontinuation.
- A blood test is required to assess FBP at cessation of therapy:
 - * If the patient is currently on weekly blood tests, weekly blood tests are required for 4 weeks post cessation.
 - * If the patient is on monthly blood tests, a single blood test is required 4 weeks post cessation.
 - * The post cessation WBC and NCs must be green (according to traffic light system) or further monitoring will be required.
- A 'Termination of Treatment Form' must be submitted to ClopineCentral™ by the treating team.

Admission of a patient currently treated with clozapine

- The ward clerk prints out the webPAS alert page and places in the front of the file with the Psychiatric Services Online Information System (PSOLIS) alert page.
- Confirm current monitoring status by contacting ClopineCentral™:
 - * If not a patient of PaRK Mental Health Service, the patient must be transferred in ClopineCentral™ from their current registered centre to the RKPG registered centre through the Mental Health ward pharmacist.
- Confirm patient compliance with therapy and the time of last dose.
- Supply of clozapine will occur on an individual patient basis after Mental Health ward pharmacist review of the patient.
 - * If the patient is deemed compliant, maintenance clozapine therapy can be prescribed on the WA Hospital Medication Chart (WAHMC) (RGMR170).
 - * The patient's CPN number, current haematological monitoring frequency (twice-weekly, weekly or monthly) and the date of next blood test must be annotated on the chart and blood test results must be entered into the ClopineCentral™ database.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			

Patients discharged from Mimidi Park

- It is the responsibility of the Treating Team Psychiatrist to ensure that the discharge summary includes the blood tests completed, results and the date next blood tests on the discharge summary and on SMHMR916 Care transfer summary to the relevant community mental health services and General Practitioner (GP).
- As per 'day patients', a copy of the Clozapine Initiation and Titration Chart (RGMR170R) will be included in the discharge information emailed to the appropriate community teams.

Protocol for GP clozapine monitoring

Refer to Appendix B: Shared Care Guidelines for Clozapine Management by GPs.

ClopineCentral™ contact details

- Phone: 1800 656 403
- Fax: 1800 657 454
- Email: ClopineCentral@pfizer.com
- Web Address: <http://www.clopine.com.au>
- ClopineCentral™ protocol forms available via this link
- The centre co-ordinator for RKPG is the Supervisor Pharmacist.

Compliance monitoring

- The Centre Coordinator is responsible for ensuring the registered centre operates in accordance with the ClopineCentral™ protocol. When contacted by ClopineCentral™ it is the Centre Co-ordinator's responsibility to investigate and resolve any overdue blood tests and incomplete documentation as outlined in this guideline.
- Compliance with this policy will be audited annually by NMHS MH Pharmacy services.
- 100% of patients will have baseline monitoring parameters and registration completed prior to commencing treatment with clozapine.
- 100% of patients treated with clozapine will be monitored in accordance with this policy.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

References

1. eTG Complete. Constipation in adults. 2016. Available from [Topic | Therapeutic Guidelines \(health.wa.gov.au\)](https://www.health.wa.gov.au)
2. Australian Commission on Safety and Quality in Health Care (ACSQH). The National Safety and Quality Health Service Standards Safety and Quality Improvement Guide: Standard 1 Governance for Safety and Quality in Health Service Organisations. 2nd ed. (Version 2) 2021. Available from <https://www.safetyandquality.gov.au/our-work/assessment-to-the-nsqhs-standards/>

Links

[Australian Commission on Safety and Quality in Health Care. National Safety and Quality Health Service Standards. Second Edition. Medication Safety](#)

[Australian Government Carer Recognition Act 2010](#)

[Australian Government. National Standards for Mental Health Services](#)

[Chief Psychiatrist's Standards for Clinical Care](#)

[DoH High Risk Medication Policy](#)

[DoH WA Health Consent to Treatment Policy](#)

[DoH WA Clinical Alert MedAlert Policy](#)

[DoH Medication Chart Policy](#)

[Health Practitioner Regulation National Law \(WA\) Act 2010](#)

[MIMS Online Clopine Full Product Information](#)

[RKPG Clinical Alerts](#)

[RKPG Medication Administration Clinical Practice Standard](#)

[Royal Perth Hospital. Clozapine e-Learning package: Therapy, side effects, monitoring requirements](#)

[The Maudsley Prescribing Guidelines in Psychiatry](#)

[WA Mental Health Act 2014](#)

[WA Medication History and Management Plan: User Guide](#)

[WA Medicines and Poisons Act 2014.](#)

[WA Medicines and Poisons Regulations 2016](#)

[WA Pharmacy Act 2010](#)

[WA Pharmacy Regulations 2010](#)

National Standards for Mental Health Services

1	2	3	4	5	6	7	8	9	10
☒	☒				☒		☒		☒

OCP Standard Clinical Care



RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix A: **Side-effects associated with clozapine therapy**

The table below outlines the more common side effects related to clozapine and signs and symptoms should be carefully monitored (outlined in the [DoH Guidelines for the Safe and Quality Use of Clozapine Therapy in the WA health system 2017](#)). For a list of pharmacological options and actions, the treating team should contact the clinical pharmacist

Side effects/Signs and symptoms	Rate	Recommended action
Haematological effects		
Neutropenia WBC < 1.5 x 10 ⁹ /L. Flu-like symptoms such as sore throat and fever may be present. (This is most common within the first 18 weeks – but may uncommonly occur after this maximal risk period).	3.2%	Requires medical review. Stop clozapine. Contact haematologist at ClopineCentral™ or clozapine patient monitoring centre if further advice required. Re-introduction of clozapine should only occur with haematologist support.
Agranulocytosis WBC < 0.5 x 10 ⁹ /L. Flu-like symptoms such as sore throat and fever may be present. (This is most common within the first 18 weeks – but may uncommonly occur after this maximal risk period).	0.8%	Requires medical review. Stop clozapine. Contact haematologist at ClopineCentral™ or clozapine patient monitoring centre if further advice required. Re-introduction of clozapine should only occur with haematologist support.
Benign ethnic neutropenia	Rare	The presence of benign ethnic neutropenia should not prevent treatment with clozapine however a haematologist review prior to commencing clozapine is of benefit which may lead to an adjustment of the white blood cell count green, yellow and red ranges.
Cardiac effects		
Myocarditis Tachycardia at rest with rapid breathing, dyspnoea, hypotension, raised jugular venous pressure, fatigue, flu-like symptoms, chest pain or fever. Myocarditis usually occurs within 1 - 4 to 6 weeks of starting).	0.7-1.2%	Requires medical review. Refer to cardiologist. Check Troponin and C-reactive protein (CRP) levels. Withhold clozapine. Repeat ECG and Echocardiogram. If confirmed contact cardiologist at clozapine monitoring centre. Submit the 'Therapy Event' form and 'TGA Adverse Event' form. Re-introduction of clozapine should only occur with cardiologist support.
Cardiomyopathy Cardiomyopathy may occur at any time. The possibility of cardiomyopathy must always be considered there is clinical evidence of heart failure including resting tachycardia, tachypnoea, shortness of breath or hypotension.	0.1%	Requires medical review. Refer to cardiologist. Check Troponin and C-reactive protein (CRP) levels. Withhold clozapine. Repeat ECG and Echocardiogram. If confirmed contact cardiologist at clozapine monitoring centre. Submit the 'Therapy Event' form and 'TGA Adverse Event' form. Re-introduction of clozapine should only occur with cardiologist support.
Tachycardia	25%	Requires medical review. Check FBC, WBC, ECG, Echocardiogram. Note that a benign tachycardia can occur in up to 25% of patients which is often self-limiting.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			

Orthostatic hypotension May be problematic for patients on clozapine therapy; it often improves as tolerance develops.	9%	Best managed with slow and gradual titration of clozapine. Patients must be provided with advice on managing postural dizziness (take time standing up) and the modification of dietary salt and fluid intake. Specialist support may be needed if the symptom persists. Treatments to address the hypotension are available (including anti-hypotensives) but must only be used under specialist support. Note that some patients may develop hypertension on clozapine.
Venous thromboembolism	Rare 0.2%	Avoid Immobilization; early detection important as potentially life threatening event.
Central nervous system effects		
Sedation First 4 weeks. This may persist but generally improves. This is the most common side effect of clozapine.	50%	Give smaller dose in the mornings. Some patients can only tolerate single night-time dosing. Reduce dose if necessary. Consider plasma level monitoring. Avoid other sedating agents.
Seizures This dose dependent risk increases with higher plasma levels (especially dosages greater than 600 mg/day), rapid dose titration, concurrent use of drugs that lower seizure threshold and pre-existing seizure disorders and illness (Seizures may occur at any time).	Up to 5%	Requires medical review. Consider reduction in dose. Clozapine need not be discontinued. Check with Clinical Pharmacist for pharmacological anticonvulsant options. Valproic Acid is commonly used due to lack of interactions with clozapine and rapid titration schedule. Monitor serum clozapine levels regularly.
Myoclonus	2%	Uncommon side effect. May be a precursor to seizures and may require an anticonvulsant.
Gastrointestinal effects		
Hypersalivation Excessive drooling – very troublesome at night. (First few months especially but can be ongoing). This is the second most common side effect.	50% (up to 85%)	May give hyoscine hydrobromide 300 micrograms sucked and swallowed at night. Hyoscine patches may be considered as an alternative. Atropine-related oral anticholinergics or sublingual ipratropium may also be considered with caution. Wrapping a towel around a pillow at night and frequent swallowing during the day can assist.
Constipation Less frequent bowel motions, hard stools, abdominal bloating, cramping or pain, decrease appetite or fatigue. (Usually persists).	14-25%	Recommend increased fluid intake and exercise. Use osmotic laxatives and, if required, stimulant laxatives.
Nausea Can develop early or later in the course of treatment.	11%	May give antiemetic. Avoid prochlorperazine and metoclopramide if previously experienced Extra Pyramidal Side Effects (EPSE).
Parotid swelling Painful swelling of the parotid glands associated with hypersalivation. In some instances, this spontaneously resolves but may persist.	Rare	Cessation of clozapine; or use of minimal effective dose. A combination of benztropine and terazosin may be considered.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			

Weight gain and metabolic effects		
Weight gain This may occur early in treatment and be can be significant. 50% Dietary and lifestyle counselling before weight gain occurs. Ongoing monitoring and support.	50%	Dietary and lifestyle counselling before weight gain occurs. Ongoing monitoring and support.
Hyperglycaemia This may occur at any time in treatment and may progress to diabetes type II.	10%	Regularly screen for evidence of diabetes. Use oral hypoglycaemics or insulin as per guidelines for managing diabetes. Consider specialist referral.
Hyperlipidaemia	Up to 50%	
Other		
Benign Fever (Temperature >38° C within first 3 weeks. * Note: a benign fever can occur in up to 20% of patients initiated on clozapine which is generally self-limiting within 3 days of onset.	Up to 20%	Contact MO. Reduce rate of dose titration of clozapine. Check FBC, WBC, troponin and CRP. Physical examination for signs of infection. Consider ECG, Echocardiogram. Do not give paracetamol until doctor notified and agranulocytosis/myocarditis excluded.
Nocturnal enuresis Loss of bladder control, especially at night (bed wetting). (May occur at any time).	23-30%	Avoid high intake of fluids in the evening and ensure adequate voiding at bedtime; plan middle-of-the-night awakenings to empty bladder. Consider the careful addition of an α adrenergic agonist / anticholinergic medication such as a tricyclic antidepressant. Desmopressin may be an effective symptomatic treatment (in nasal or oral form). Consider a continence referral.
Obsessive Compulsive Behaviour Symptoms may be transient but can follow a more persistent and chronic course and can be disabling.	10-40%	Dose reduction may lead to symptom improvement. Cognitive behavioural therapy or pharmacological agents such as serotonin-specific reuptake inhibitors (antidepressants). Caution must be applied as many antidepressants can have an effect on clozapine serum levels.
This is not an exhaustive list of side-effects. Please see Clozapine Product Information for further advice.		

Referenced from [DoH Guidelines for the Safe and Quality Use of Clozapine Therapy in the WA health system 2017](#)

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			

Appendix B: **Shared care guidelines for clozapine management by GPs**

Scope (Staff): Peel and Rockingham Kwinana Mental Health Service (PaRK MHS)
Staff: Nursing/Medical
Scope (Area): PaRK MHS Community Mental Health Clinics

Aim

To provide an evidenced based guideline for PaRK MHS Community Mental Health Clinics (CMHC) to manage the transition of care of stable clozapine patients to their GP. This guideline is recommended to GPs who are taking over the care of patients on clozapine. This guideline is based on best practice and allows for the safe and effective use of clozapine, including meeting the patient's physical health needs and recovery focussed care in the community.

Scope

This outlined procedure provides information for all clinicians involved in the care of GP managed clozapine patients.

Background

- Clozapine has proven clinical efficacy and a strong research background with the clinical benefits to patients empirically acknowledged including treatment for resistant schizophrenia. However, clozapine can cause potentially life threatening side effects. Therefore, ongoing haematological and physical health care monitoring is required for all patients on clozapine.
- To ensure the effective management of clozapine patients [Guidelines for the Safe and Quality Use of Clozapine in the WA health system.pdf](#) states that:

"A number of potential service models providing improved integration of mental and physical health care have been explored in other jurisdictions. The importance of the general practitioner in all aspects of care, and their potential for providing leadership is emphasised."

"The particular approach selected will be determined largely by patient characteristics, modified by resource availability, and more than one model may be used within a service."
- PaRK MHS CMHCs have identified a cohort of patients who would find substantial benefit to having their complete care managed by their GP. These patients have been stable for a period of 12 months or longer and take ownership of their own care attending for monthly bloods and medical appointments without any assistance.
- The Guidelines for the Safe and Quality Use of Clozapine Therapy in the WA health system state that the following considerations need to be accounted for before engaging in a GP clozapine shared care arrangement.

Considerations for clozapine shared care

- Strong links must be established between all parties in clozapine shared care models. It is important to highlight that all models of shared care for the management of clozapine must ensure the safety of patients, that there is no loss of quality of

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

service in the transition to shared care, that all clinicians involved are competent and qualified professionals, and that specialist oversight of the care remains (as is required both legislatively and clinically). All of the community professionals involved in the prescribing and dispensing of clozapine must be appropriately educated and registered and must have retained clear links with clozapine coordinators and shared care coordinators either in the public or private health systems.

- Not all clozapine patients on maintenance therapy are suitable for community prescribing or dispensing. It is the characteristic of the patient receiving the clozapine rather than the duration of clozapine therapy which must be the determining factor in any transition of care.
- Factors that may impact on a patient's ability to move to community based care include the following:
 - * A patient's compliance history with clozapine and other medication.
 - * Their ability to independently attend appointments, blood tests and other investigations.
 - * Long-term sustainable support (family, partner, carer).
 - * Their ability to access a suitable Pharmacy.
 - * Their satisfaction and consent for transition to community care.
 - * Their practical ability to access the community scheme.
- Any transition of a patient's clozapine management out of the traditional hospital or community clinic based model requires careful planning, preparation and monitoring to ensure sustained success. All services must have a full understanding of legislative or other requirements before implementing new models of care for clozapine. Where public mental health services retain involvement in the care of a patient on clozapine under any model of care, the service must continue oversight of any shared care models.
- This guideline incorporates the considerations outlined by "The Guidelines for the Safe and Quality Use of Clozapine Therapy in the WA health system" to ensure best practice patient care.
- The final decision regarding the patient's suitability to move to community based care is made by the Consultant Psychiatrist.

Procedure for discharge to GP

- Identification of patients to be considered eligible for discharge to the care of their GP for clozapine prescribing; a patient must meet the following criteria:
 1. Have been on clozapine for at least 12 months.
 2. Be stable on 4-weekly blood tests.
 3. Clinically stable with no mental health admissions in the previous last 12 months
 4. Be self-managing with regard to attending for blood tests, appointments and taking medication (blister packs acceptable).
 5. Patients that adhere to the requirements of the treatment with support from family or supported accommodation may also be deemed suitable if the supports, treating and GP are in agreement.
 6. No/Low case management needs.
 7. Stable dose of psychotropic medication for 12 months.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			

8. Patient and GP are agreeable to transition.
9. Consideration of financial implications in particular for clients that do not hold a concession card.
10. Risks to be specifically addressed (considerations but not absolute exclusions):
 - a. History of overdose
 - b. Significant current illicit drug use
 - c. Recent history of red result within the previous 12 months of taking clozapine (WBC < 3.0 and NC < 1.5)
 - d. Significant non-adherence
- All the following considerations are assessed on a case by case basis with other considerations not stated also possible based on a patient's circumstances.
- Once a patient has been identified as suitable for discharge to the GP the case manager or GP Liaison Officer will liaise with the patient and provide supports to the GP to facilitate the transfer of care. A handover document will be created for the GP detailing the relevant history e.g. physical health monitoring results, recent medical reviews, current medication list, current Pharmacy and any other relevant information specific to the patient. The GP Liaison Officer or case manager may attend the first appointment to provide further handover. Appendix C is a template letter that the treating doctor may wish to use.

GP

- For a GP to take over the clozapine management of their patient, the GP must first agree undertake the following measures:
 - * Register with ClopineCentral.
 - * Adhere to Clopine policies and procedures.
 - * Prepare a GP mental health care plan at initial consultation.
 - * Ensure cover if on leave with a Clopine registered GP.
 - * Willing to bulk bill.
 - * Referral for psychiatrist review every 12 months as required for PBS prescribing.
- The GP must also be willing to refer back to or liaise with the relevant CMHC if required due to:
 - * Blood tests are in amber or red range **WCC <3.5 or NC <2.0** **WCC <3.0 or NC <1.5**.
 - * There has been a therapy interruption of greater than 2 days.
 - * Significant deterioration in mental state.
 - * Patient/carer requests review.
 - * Evidence of non- adherence with medication, blood testing or not attending appointments.
 - * Unable to cover a clozapine appointment with a replacement GP in an acceptable time frame to meet clozapine protocol.
 - * Pregnancy.
 - * Clozapine dose change.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			

- **Immediate referral to the clinic** if clozapine needs to be ceased for any reason including red results.
- **Dose changes of clozapine:** it is important that the GP consults with the CMHC psychiatrist whenever they believe a dose adjustment is indicated.

First appointment

The first appointment between the patient and their GP for clozapine management may be attended by the GP liaison officer or case manager, to ensure that the clinical handover has occurred, to give the patient and GP any further information required, and to ensure that the patient is booked in for their monthly appointments. It is expected that the GP will create a GP mental health plan at this appointment.

Subsequent appointments

(Must be performed every four weeks)

- Assess for:
 - * Any change in bowel habits and ensure no constipation.
 - * Any symptoms that may suggest the onset of cardiac illness including tachycardia, chest pain or shortness of breath (potential signs/symptoms of clozapine-induced cardiomyopathy/myocarditis).
 - * Any flu like symptoms including fever, malaise and myalgia (potential signs/symptoms of clozapine-induced neutropenia).
 - * Check for drug interactions.
 - * Ask about side effects.
 - * Review caffeine consumption and cigarette smoking habits.
 - * Review of mental health.
 - * Review of adherence.
- Physical health assessment as required and 6 monthly as part of metabolic screening:
 - * BP
 - * HR
 - * Temperature
 - * Weight
- **Note:** Appendix D is a form that GP's may wish to use to document this information and guide the assessment. Appendices H and J are provided to help with assessing side effects and constipation. Appendices G, K, L, M and N have been provided as a resource tools.

Physical health monitoring

(To be performed when required)

Due to the effects of clozapine, ongoing physical health monitoring and management is essential. It is recommended that ongoing physical health monitoring is provided as per the management plan provided (Appendix E is a template)

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Prescribing clozapine

- Clozapine can only be supplied until the date of the patient's next blood test. Therefore, clozapine should be prescribed in the quantity required to ensure the patient has enough medication until their next blood test only (usually 28 days).
Note: the maximum PBS quantity that can be prescribed which is 200 tablets per strength.
- Clozapine can be prescribed at each appointment with the streamline code or it is possible to obtain phone authority to prescribe up to five repeats for monthly dispensing to reduce the need to prescribe at each appointment. The Pharmacy will not dispense without a current blood test signed off by the doctor and will usually retain the repeats.
Note: Clopine Brand must be prescribed for PaRK MHS patients as this is the monitoring system used (no brand substitution).

ClopineCentral

- For a clozapine patient whose clozapine is being managed by their GP the Clopine Central set up will be as follows:
 - * CMHC will be the overseeing "Clopine Centre"
 - * GP will be registered as a "Clinic" of the relevant CMHC Clopine Centre
 - * Community Pharmacy dispensing clozapine will also be registered as a "Clinic" under the relevant CMHC Clopine centre as is currently the case for Clopine dispensing pharmacies in the PaRK Catchment.
- This set up will allow the GP Liaison Nurse of the relevant CMHC to identify if either a GP or Pharmacy is having any difficulty with the Clopine process and assist follow up. It will also allow the GP and Pharmacy to have access only to the information of the patients they are responsible. This set up will also make it simpler for the CMHC to access patient information should consultations with the GP be needed and to allow simple transfer if management needs to be taken back by the CMHC due to any issues encountered.

Community mental health clinic review

- Every 12 months the GP will refer their clozapine patients to the relevant CMHC for their annual review with a psychiatrist. The purpose of this review is to maintain a working relationship between the clinic, patient and their GP and provide the oversight required for the PBS prescribing of clozapine in the community.
- The GP Liaison Nurse will oversee patients discharged on this program, monitor Clopine alerts and be a point of contact for any queries or concerns.

Roles and responsibilities

Non-attendance for appointments

If a patient does not attend an appointment as required the following steps should be taken first by the GP surgery and then if required by the community mental health clinic to which the patient is attached.

Step 1

- GP practice should attempt to contact the patient to arrange another appointment

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

- If unable to contact patient attempt to contact nominated support person.
- If able to locate the patient and they are able to attend an appointment and receive a blood test without a therapy interruption of >2 days, clozapine management can continue as usual.
- If able to locate the patient however, there has been a therapy interruption of over 2 days, the GP will contact the nominated Psychiatrist or GP Liaison Nurse at the CMHC and consult on re-titration of clozapine.

Step 2

- If the patient has not been assessed by the GP within one week of the due date for the required clozapine blood test, the GP will contact the CMHC to inform them that the patient has not attended and the CMHC will follow up the patient.

Step 3

- CMHC will reengage the clozapine patient and attempt to address any issues affecting the patient with the aim of providing support to help the patient to continue on clozapine.
- CMHC will liaise with the GP and the patient on whether the CMHC will take back management of patient's clozapine or return management to the GP on a case by case basis.

No blood test done/No result

Step 1

- Attempt to identify the cause/barriers of the patient not receiving a blood test or why the results are not back. If satisfied that the patient will have their blood test the GP can continue the clozapine assessment as usual. However, consideration must be given to the number of days' medication the patient has available. In these cases, a clozapine script should not be written or the dispensing of a repeat authorised until the patient receives a blood test and it is reviewed by the GP.
- If the patient will run out of medication before a blood test can be done and reviewed and the GP is comfortable with their assessment, the GP will contact ClopineCentral to allow a short supply of medication to be prescribed until the patient can receive the blood test and the GP can review it. Refer to the ClopineCentral Protocol.
- **Note:** If the patient does receive a late blood test after their monthly appointment and the GP is confident there has not been an interruption to therapy of more than 2 days, after reviewing the bloods a prescription can be written and blood results and prescription can be forwarded to the nominated Pharmacy.

Step 2

- If the patient has not received a blood test **within one week** of the due date, the GP will contact the clinic to inform them that the patient has not received a blood test or more medication and the clinic will follow up the patient.

Step 3

- CMHC will reengage the clozapine patient and attempt to address any issues affecting the patient with the aim of providing support to help the patient to continue

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			

on clozapine or change treatment plan.

- CMHC will liaise with the GP and the patient on whether the CMHC will take back management of patient's clozapine or return management to the GP on a case by case basis.

Non-compliant with clozapine

Step 1

- The GP should contact the relevant CMHC to discuss this with a clinician for a consult and future management.

Step 2

- The CMHC will re-engage the clozapine patient and attempt to address any issues affecting the patient with the aim of providing support to help the patient to continue on clozapine or change treatment plan.
- CMHC will liaise with the GP and the patient on whether the CMHC will take back management of patient's clozapine or return management to the GP on a case by case basis.

Psychiatrist

- A consultant psychiatrist or senior medical practitioner will be available at each community mental health clinic during the working week for the GP to discuss any issues relevant to clozapine patients being managed by them. Each mental health clinic will have a GP Liaison Nurse that will triage calls from GP's and where required will organise urgent contact with a consultant psychiatrist for review and advice. The Assessment and Treatment Team (ATT) will be available for advice in an after-hours emergency.

GP Liaison Officer

- Each community mental health clinic will have a GP Liaison Nurse who will maintain oversight of GP-managed clozapine patients through the administration of the ClopineCentral database. The GP Liaison Nurse will also be available to provide support, including but not limited to:
 - * Triage calls from GP's, involving team psychiatrist where required.
 - * Providing education and support on clozapine protocol to GP's and community Pharmacists.
 - * Managing Clopine registration of new personnel, GP's and Pharmacies on the Clopine Central site.
 - * Attend patient appointments if required, where possible.
 - * Support, assist and provide troubleshooting with issues that may arise in patient management.

Community Pharmacy/Pharmacist

- The process of transfer of care to the GP will include identification of the patient's preferred community Pharmacy as this may change when the patient no longer needs to attend the clinic for clozapine management.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

- The community Pharmacy will be advised of the transfer of care prior to the first GP appointment to ensure the required registrations are in place. The community Pharmacy must:
 - * Be registered with ClopineCentral.
 - * Adhere to Clopine policies and procedures.
 - * Only dispense clozapine once reviewed blood results have been entered/viewed on ClopineCentral.
 - * Liaise with GP as needed.
 - * Attempt to contact the patient if the patient does not collect clozapine when required. If not able to make contact with the patient the pharmacist will contact the GP to discuss.

Clozapine patient

For transfer of clozapine to the care of their GP the patient must agree to actively participate in:

- Attending a pathology centre for blood test the morning or day before their appointment with their GP (every 28 days or as required by protocol)
- Attend appointment with GP every 28 days – or as required by protocol
- Attend their nominated Pharmacy with blood count form and collect medication
- Attend review appointment at community mental health clinic every 12 months
- Contact GP if any concerns in between appointments particularly constipation, sore throat or feeling unwell
- Attend GP for metabolic screening as required by protocol

Centre Coordinator

The PaRK Centre Coordinator for Clopine will monitor:

- Overdue blood (contact community Pharmacy and escalate to case manager/GP liaison Officer).
- Abnormal blood result i.e. ambers, red (contact case manager or GP liaison Officer).
- therapy events (advise treating team of outstanding case that need review).

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Contact Information

PaRK MHS CMHC Contact information

Rockingham Kwinana Mental Health Service

Corner of Ameer and Clifton Street

Rockingham 6168

Phone (08) 9528 0600

Fax (08) 9529 1266

Operating Hours Monday to Friday 08:30 - 16:30

Peel Community Mental Health Service

110 Lakes Road

Mandurah 6210

Phone (08) 9531 8080

Fax (08) 9531 8070

Operating Hours Monday to Friday 08:30 - 16:30

Pharmacy Rockingham General Hospital

Telephone: 9599 4000

Operating hours: Monday to Friday 08:00 – 16:00

Clopine Connect

Telephone: 1800 656 403

clopinecentral@pfizer.com

Mental Health Emergency Response Line (MHERL) Assessment Treatment Team After Hours Until 22:00.

Telephone: 1300 555 788

Operating Hours: 24/7

State-wide Psychotropic Drug Information Centre

Telephone: (08) 9347 6400

Email: DrugInformation.Graylands@health.wa.gov.au

Operating Hours Monday to Friday 08:30 - 16:30

After Hours On-Call Pharmacist and DMO Rockingham Hospital Switchboard: 9599 4000 Operating Hours Monday to Friday 16:30 -08:30, Saturday, Sunday and Public Holidays 24 hours

WA Poisons Information Centre

Telephone: 13 11 26

Operating Hours 24/7

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix C: **Clozapine management by GP** (Template letter the treating doctor may wish to use)

DATE

GP details

CLOZAPINE PATIENT

Dear Dr.....,

Re: Mr Paul Patient (DOB 16/4/1956)
UMRN 000000
Clopine number:

Thank you for accepting **NAME** as part of the Clozapine Shared Care Programme. **NAME** has now been discharged from our service due to ongoing stability of both their mental state and medication regime. Under the Medicare requirements for PBS prescribing of clozapine, **NAME** needs to be seen every 4 weeks by yourself at the above practice (or by a Clopine registered GP) for their clozapine monitoring and physical health management. Our service will review every 12 months at our clinic or more frequently if required. As you would be aware, Clozapine can present risks to physical health, including life-threatening adverse effects, and as such requires close monitoring.

Side effects include constipation, neutropenia, cardiac effects such as myocarditis, cardiomyopathy and tachycardia, over sedation, hypersalivation, hypotension, and metabolic syndrome.

Attached to this letter for your information is a table of recommended physical health monitoring for a patient taking Clozapine. Below are also links to useful information for the management of patients on clozapine and their physical health requirements. These include a very informative guide to clozapine in primary care provided by Australian Prescriber as well as customer information that can be accessed at the bottom of the document 'Guidelines for the Safe and Quality Use of Clozapine Therapy in the WA Health System'.

Clozapine has many important drug interactions that may significantly affect its plasma levels including stopping or starting smoking, some antidepressant medications as well as other potent inhibitors of cytochrome P450 isoenzymes. Significant changes in plasma clozapine levels may manifest as changes in mental state or changes in side effects such as sedation and hypersalivation. Other medications that can cause neutropenia such as carbamazepine or immunosuppressant's may need to be closely monitored if given concomitantly.

If the patient reports the onset of any signs or symptoms of either cardiac illness or infection potentially worsened by neutropenia contact our team or the Clopine Patient Monitoring Service on 1800 656 403 immediately as further blood tests/monitoring may be required.

Should you have any questions or concerns with this patient or their Clozapine treatment, please do not hesitate to contact either Dr..... orCommunity Treatment Team.

Current Medications From Clinic

Current Physical measurements

BP, Weight, Waist circumference

Other important measures/Information

WCC, NEU, BGL

Yours sincerely

DR details

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix C contd: **GP checklist for monthly appointments**

Mental State examination:- Appearance, Behaviour, Speech, Mood, (Any changes or concerns) Psychotic symptoms - (Any changes or concerns) Risk of harm to self/others – (suicidal/self-harm or homicidal)	
Bloods taken in last 24 hours:- Green – WBC >3.5 Neutrophils >2.0 Amber – WBC 3.0-3.5 Neutrophils 1.5-2.0 Red – WBC <3.0 and/or Neutrophils <1.5 Seek advice from Mental Health Service if Amber or Red.	
Screen for side effects:- Constipation – refer to Appendix H	
Any changes to smoking, request serum clozapine level.	
Complete Blood record form and prescription for clozapine.	
Ensure next appointment is made in 4 weeks' time and blood form given.	
If client does not attend:- Attempt contact if client. Contact support person if not able to reach client. Check no interruption of therapy longer than 2 days. Contact Mental Health Service if needed.	
GP Liaison Nurse Rockingham Mental Health Service Duty Officer Rockingham Mental Health Service GP Liaison Nurse Peel Mental Health Service Duty Officer Peel Mental Health Service Clopine Support Line After Hours Community Mental Health Team For on call Pharmacy or duty Dr at Rockingham Hospital	<u>9528 0600</u> <u>9528 0600</u> <u>9531 8080</u> <u>9531 8080</u> <u>1800 656 403</u> <u>1300 555 788</u> <u>9599 4000</u>

Clozapine Management by GP

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix D: **Management plan template for subcut clozapine**
(copy to be given to GP, client, support person and for the file)

Brief overview of client:-

Diagnosis:-

Current medication and duration:-

Medication compliance (include side effects):-

Mental Health History:-

Drug and alcohol issues:-

Previous medication (and reasons for ceasing):-

Hospital admissions:-

Relapse warning signs:-

Action (steps to be taken, coping strategies)

Ongoing medical monitoring:-

LFTS 6 monthly. Last completed results next due.....

Serum Clozapine level 6 monthly. Last completed results next due.....

Fasting blood glucose and lipids 6 monthly. Last completed results next due.....

Troponin and CRP 6 monthly. Last completed results next due.....

Metabolic screening 6 monthly including weight, waist circumference, BMI, blood pressure, heart rate.

Last measurements:-

ECG annually. Last completed results next due.....

Echocardiogram (year 1, 2, 5, and 10). Last completed results next due.....

Support details (family/friends etc):-

GP details:-

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix E: **Checklist for clozapine GPSC transition – file copy**

Confirmed that client has been stable 12/12 on clozapine, all green blood results and established regular contact with GP.	
Care Plan with all relevant information completed and discussed with GPL Nurse.	
Medical letter from last review with medical staff at PARK.	
Clozapine Level, CRP and troponin 6 monthly Metabolic bloods 6 monthly <u>Next due</u>	
Latest ECG (12 monthly) date and results:- <u>Next due</u>	
Latest echocardiogram (due at year 1,2,5 and 10) date and results:- <u>Next due</u>	
Identified Pharmacy	
Identified Pathology	
Letter from GPL Nurse/CM re plan.	
Next Psychiatrist appoint at clinic in 12 months	
Support Person Informed and involved if applicable.	

PaRK ONLY

PSOLIS personal support person and GP details updated:

SMS reminder for 12 month appoints:

Alert entered in PSOLIS and file stating on GPSCP:

Add to discharged list maintained by GPL Nurse -

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix F: **Non metabolic screening protocol for patients on clozapine**

Date Commenced:														
Date	FBP	Troponin	CRP	ECG	LFT	U+E	Echocardiogram	AIMS	Prolactin	βHCG	Serum Clozapine (write value)	Blood Group		
													6 mths	Baseline
													12 mths	1 wk
													18 mths	2 wks
													2 yrs	3 wks
													2.5 yrs	4 wks
													3 yrs	5 wks
													3.5 yrs	6 wks
													4 yrs	7 wks
													4.5 yrs	8 wks
													5 yrs	9 wks
													5.5 yrs	10 wks
													6 yrs	11 wks
													6.5 yrs	12 wks
													7 yrs	13 wks
													7.5 yrs	14 wks
													8 yrs	15 wks
													8.5 yrs	16 wks
													9 yrs	17 wks
													9.5 yrs	18 wks
													10 yrs	19 wks monthly onward

Non metabolic screening protocol for patients on Clozapine
Indicate **N** (normal) or **A/N** (abnormal) in relevant white box once completed (grey boxes are not mandatory but repeat any tests when required)

RESPONSIBILITY				NATIONAL STANDARDS	
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.	
This version	Dec 2022	Revision Due	Dec 2025		
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee		
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure				

Appendix G: Important drug interactions with clozapine

Drug	Interactions	Comments
CYP1A2 inhibiting substances (e.g. fluvoxamine, caffeine, ciprofloxacin)	Concomitant use may increase clozapine levels	Potential for increase in plasma clozapine levels and adverse effects. Care is also required upon cessation of concomitant CYP1A2 inhibiting medications as there will be a decrease in clozapine levels.
CYP2D6 inhibiting substances (e.g. fluoxetine, paroxetine)	Concomitant use may increase clozapine levels	Potential for increase in plasma clozapine levels adverse effects. Care is also required upon cessation of concomitant CYP2D6 inhibiting medications as there will be a decrease in clozapine levels.
Bone marrow suppressants • carbamazepine, - chloramphenicol • sulphonamides (e.g. co-trimoxazole), • pyrazolone analgesics (e.g. phenylbutazone), • penicillamine, • cytotoxic agents and • long-acting depot injections of some antipsychotics	Interact to increase the risk and/or severity of bone marrow suppression.	Clozapine use is not recommended concomitantly with other agents having a well-known potential to suppress bone marrow function. Depots have been used clinically in some cases.
Anticholinergics	Clozapine potentiates action of these agents through additive anticholinergic activity.	Observe patients for anticholinergic side-effects, e.g. constipation, especially when using to help control hypersalivation.
Antihypertensives	Clozapine can potentiate the hypotensive effects of these drugs due to its sympathomimetic antagonistic effects.	Caution is advised due to potentiation of hypotensive effects, especially during the period of initial dose titration.
Alcohol, MAOIs, CNS depressants, including narcotics and benzodiazepines	Enhanced central effects. Additive CNS depression and cognitive and motor performance interference when used in combination with these substances.	Caution is advised if clozapine is used concomitantly with other CNS active agents. Advise patients of the possible additive sedative effects and caution them not to drive or operate machinery.
Highly protein bound substances (e.g. warfarin and digoxin)	Clozapine may cause increase in plasma concentration of these substances due to displacement from plasma proteins.	Patients should be monitored for the occurrence of side effects associated with these substances, and doses of the

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

		protein bound substance adjusted, if required.
Phenytoin	Addition of phenytoin to clozapine regimen may cause a decrease in the clozapine plasma concentrations.	If phenytoin must be used, the patient should be monitored closely for a worsening or recurrence of psychotic symptoms.
Lithium	Concomitant use can increase the risk of development of toxicity.	Observe for signs and symptoms of toxicity.
Drug	Interactions	Comments
Antibiotics and Erythromycin	Use of antibiotics may produce a rogue low neutrophil count on blood sampling. Erythromycin may exacerbate the side effects of clozapine due to increasing plasma levels	Use of antibiotics due to clinical need should be reported to the local Clozapine Clinic. Observe for increase side effects with erythromycin.
CYP1A2 inducing substances (e.g. carbamazepine)	Concomitant use may decrease clozapine levels	Potential for reduced efficacy of clozapine should be considered.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix H: **Constipation assessment tool**

The normal frequency of bowel motions can vary between individuals from 3 times per day to 3 times per week.¹

Therefore, constipation should not be diagnosed based on the frequency of bowel motions alone.¹

Constipation should be assessed using the following criteria in conjunction with the Bristol Stool chart. Constipation can be confirmed when 2 or more of the following are present.¹

- ☐ two or fewer motions per week
- ☐ straining at more than 25% of defecations
- ☐ a feeling of incomplete evacuation at 25% or more defecations
- ☐ hard lumpy stools (1 or 2 on the Bristol Stool Chart)
- ☐ sensation of anorectal obstruction or blockage for at least 25% of defecations
- ☐ manual manoeuvres needed to facilitate at least 25% of defecations (e.g. digital evacuation, support of the pelvic floor).¹

It is important to review current medication therapy and ask about the use of over the counter medications, complementary medicines and laxatives to assist in identifying possible causes of constipation and determine appropriate management.¹

Other common issues that could be exacerbating or causing constipation should also be assessed including:

- ☐ review other medication that could be causing constipation (particularly anticholinergics)
- ☐ inadequate dietary fibre
- ☐ dehydration (e.g. not drinking enough water)
- ☐ inappropriate bowel habits (e.g. ignoring the urge to defecate)
- ☐ inadequate physical activity (e.g. from decreased mobility)
- ☐ change in environment (e.g. holiday, entry into a residential aged care facility)
- ☐ painful anorectal disorders (e.g. haemorrhoids, anal fissure)
- ☐ loss of muscle power caused by other medical conditions (e.g. chronic obstructive pulmonary disease).¹

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix I:

Bristol Stool Chart

Type 1		Separate hard lumps, like nuts (hard to pass)
Type 2		Sausage-shaped but lumpy
Type 3		Like a sausage but with cracks on the surface
Type 4		Like a sausage or snake, smooth and soft
Type 5		Soft blobs with clear-cut edges
Type 6		Fluffy pieces with ragged edges, a mushy stool
Type 7		Watery, no solid pieces. Entirely Liquid

First published: Lewis SJ, Heaton KW (1997) Stool form scale as a useful guide to intestinal transit time. Scandinavian Journal of Gastroenterology 32: 920–4

Type 1–2 indicate constipation Medical team should review and provide treatment

Type 3–4 are ideal stools as they are easier to pass, and

Type 5–7 may indicate diarrhoea and urgency. Review and if indicated there may be the need to cease/reduce laxatives or consider symptoms as a sign of overflow.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			

Appendix J: **GASS for clozapine**

Name: _____

Current Medications: _____

Date: _____

Caffeine intake:cups/day

Smoker: Y / Ncigarettes/day

Has there been a recent change in your smoking habit?: Increase/Decrease by.....
cigarettes/day.

This questionnaire is being used to determine if you are suffering from excessive side effects from your medication. Please put a tick in the column which best indicates how often or how severely you have experienced the following side effects.

Over the past week:	Never	Once	A few times	Everyday	Tick if severe or distressing
1. I felt sleepy during the day					
2. I felt drugged or like a zombie					
3. I felt dizzy when I stood up or have fainted					
4. I have felt my heart beating unusually fast or irregularly					
5. I have experienced jerking limbs or muscles					
6. I have been drooling					
7. My vision has been blurry					
8. My mouth has been dry					
9. I have felt sick (nauseous) or have vomited					
10. I have felt gastric reflux or heartburn					
11. I have had problems opening my bowels (constipation)					
12. I have wet the bed					
13. I have been passing urine more often					
14. I have been thirsty					
15. I have felt more hungry than usual					
16. I have been gaining weight					

I have also experienced:

(please write down any other side effects that you may have experienced over the past week)

16	
17	
18	

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix K:

Staff Information

1. Allow the service user to fill in the side-effects scale themselves. All questions relate to the previous week.

2. Scoring

0 Points	"Never"
1 point	"Once"
2 points	"A few times"
3 points	"Everyday"

3. Results

0-16	absent/mild side-effects
17-32	moderate side-effects
33-48	severe side-effects

4. Side-effects covered include:

1-2	Drowsiness and sedation
3	Postural hypotension
4	Tachycardia
5	Myoclonus
6	Hypersalivation
7-8	Anticholinergic side-effects
9-10	Gastrointestinal side-effects
11	Constipation
12	Nocturnal enuresis
13-14	Screening for diabetes mellitus
15-16	Weight gain

5. The column relating to the severity/distress experienced with a particular side effect is not scored, but is intended to inform the clinician of the service user's views and condition.

Adapted from the Glasgow Antipsychotic Side-effect Scale® 2007 by St. John of God Hospital and South London and Maudsley Trust

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix L: Fact Sheet – Getting an amber blood result



Government of Western Australia
Department of Health



Handy Fact Sheet Clozapine - getting an amber blood result

As you know regular blood testing is needed when you are taking clozapine. This is because clozapine can cause a drop in the number of your white blood cells. White blood cells are made in the bone marrow and are used by the body to fight infections. If you don't have many white cells, your body will find it hard to fight infections such as flu or a sore throat.

The traffic light system

The clozapine system uses a traffic light coding to rank the results from your blood tests. The system uses a 'red', 'amber' and a 'green' light system:

Red means your white blood cells count is much lower than normal. You would need to stop taking clozapine straight away.



Amber means your white blood cell count is a little lower than normal. You will need a blood test twice a week until it goes back to normal ('green') or goes 'red'.

Green means your white cell count is normal and you can carry on as normal.

What causes an amber result?

An amber result occurs in about 1 in 50 people (2%), often in the first three months of taking clozapine, but it can occur anytime.

It can be caused by, or made worse by, many different things other than clozapine, such as:

- Other medicines that can affect the bone marrow. You must let your doctor know of

any other medication that you might be taking e.g. antibiotics

- Your health e.g. having 'flu
- The time of the day — white cell counts are lower earlier the day
- Your ethnicity (e.g. it seems to be more likely if you are Asian or Afro-Caribbean)
- Chance — everyone's white cell count goes up and down and sometimes it goes below the clozapine limits
- It can also be a blood testing error.

What happens now I've got an amber result?

- Firstly, you will have another blood test to make sure it wasn't an error
- If it wasn't you must have a blood test twice a week until your result either goes 'red' or 'green'.
 - If it goes 'red', you will have to stop clozapine
 - If it goes 'green' you can go back to your usual blood testing frequency.

You can carry on taking clozapine unless your blood goes 'red'.

If you get any 'flu-like symptoms such as a sore throat or fever let your doctor know as soon as possible.

☎ Lifeline provides 24hr telephone crisis support on 13 11 14 or visit www.lifeline.org.au for information & downloads

The small print: This leaflet is to help you understand more about getting an amber blood result with clozapine. You should also read the manufacturer's Consumer Medicine Information (CMI) Leaflet.

V08.02 L14 [SRB 11.2020] ©2020 Mistura™ Enterprise Ltd (www.choiceandmedication.org). Choice and Medication™ indemnity applies only to licensed subscribing organisations and the personal use by that organisation's service users and carers. Use by non-subscribing organisations is prohibited.

<https://www.choiceandmedication.org/wadoh/>

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix M: Fact Sheet – what carers need to know



Government of Western Australia
Department of Health

Handy Fact Sheet Clozapine – what carers need to know

This leaflet complements the Choice and Medication Information Leaflets and Handy Fact Sheets for clozapine.

This fact sheet has been written to try to help you (as a carer, friend, relative, or partner) to help someone taking clozapine to carry on recovering safely and then to stay well. This is because clozapine can be really effective but sadly does have a wide range of side effects, many quite unpleasant and some dangerous.

Clozapine (also called Clozaril® and Clopine® in Australia) is used to help treat the symptoms of schizophrenia in people who have not done well on at least two other antipsychotics e.g. not helped or who have had bad side effects (often called Treatment-Resistant Schizophrenia). Clozapine can also be used for the psychosis that can occur in Parkinson's Disease, mainly caused by the medicines to treat the symptoms.

Your local mental health service will usually prescribe and supply clozapine. But, once someone is at home or in a caring facility, there are many things you can do to help the person get the best out of clozapine, keep on their road to recovery and then to stay well.

The main important factors with clozapine		
What the issue is:	Why it is important:	What carers can do to support the person:
The person needs to take clozapine every day	- If they don't take clozapine every day, they are much more likely to become unwell again - If they miss taking clozapine for more than 48 hours it is not safe to re-start on their usual dose - If they start again at their usual dose they may get a sudden and severe drop in blood pressure, which can be harmful.	- Carers can support them to take their clozapine every day as prescribed - If the person is struggling to take their clozapine every day, you can explore with them what the problem is - There are a range of "compliance aids" that can help people remember to take a dose, and to know if they have already have taken it - Their pharmacy may be able to help or they may need talk about side effects with their mental health team.
		If the person has not taken clozapine for 48 hours or more, contact their mental health team. Clozapine will have to be re-started from a low dose over several weeks.
The person needs to have regular blood tests (full blood counts or FBC): - Weekly for the first 18 weeks - Every 4 weeks from week 19 onwards.	- Clozapine can cause the number of white blood cells to drop, in about 1 in 100 people - This puts them more at risk of getting ill from an infection - Having regular blood monitoring makes clozapine very safe.	- You can support the person to have their blood tests at the right time; they can't be given the next supply of clozapine unless they've had a blood test - Watch out for signs of infection which might mean that their white blood cell numbers have dropped - If they get a sore throat, 'flu or a high temperature contact their GP or a member of their mental health team as they might need an extra blood test.

Page 1/4

<https://www.choiceandmedication.org/wadoh/>

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8. OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			



Some other problems with clozapine		
What the issue is:	Why it is important:	What carers can do to support the person:
Being physically unwell may affect clozapine treatment	- Being unwell with an infection can make clozapine levels go up a bit and this can cause more side effects - The clozapine dose might need to be reduced a little during this time 'Flu-like symptoms can be a sign of more serious side effects - Keeping hydrated and cool is very important to maintain the body.	- Carers can help the person contact their mental health team if they become physically unwell (especially if they have a chest infection) - Carers need to be aware of the signs that someone's clozapine level is going too high e.g. being extra sleepy, having a fit or seizure, feeling dizzy, falling over - Wearing the right clothes in hot weather is important e.g. <i>not</i> big jumpers - Alert the person's mental health team if these happen.
There is still a chance the person may become mentally unwell again even though still taking clozapine	- Clozapine is the most effective antipsychotic, but it doesn't work fully for everyone - Other things may happen which lead to a relapse in the person's mental health.	- Carers should know the signs of someone becoming unwell again - They may not notice these signs themselves - Contact their mental health team in the next few days if this starts to happen.
The GP or other healthcare professionals (HCPs) may not know the person is taking clozapine	- Clozapine is started by Psychiatrists, and prescribing and supply usually remain with a mental health service - Because of this, the GP practice and a community pharmacy might not always know the person takes clozapine because they don't prescribe or supply it Clozapine needs to be on their GP records so that: - They are not accidentally prescribed any medicines that interfere with clozapine - The HCP knows if the person becomes unwell that clozapine may be a part of the cause.	- Carers can support the person to make sure their GP has recorded that they are prescribed clozapine from another source - Make sure they tell any pharmacist, doctor, nurse, mental health worker or dentist who treats the person that they are taking clozapine.
Stopping clozapine	- There might come a time where the person feels it is the right time to try to stop clozapine - It is important to know how to do this safely to cut the risk of becoming unwell again or without making symptoms worse. It is generally much safer to stop slowly than quickly - Having serious side effects or high blood levels may mean stopping more quickly but this needs support.	- It may be useful to talk with other carers about the pros and cons of stopping - Make sure prescribers and healthcare professionals know and are fully informed if someone decides to try to come off clozapine - Clozapine may need to be stopped for a few days – but this must only be done with medical advice and support. See our Handy Fact Sheets on 'Stopping or Coming Off Medicines', which may help them make their decisions.

Page 2/4

<https://www.choiceandmedication.org/wadoh/>

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.



What the issue is:	Why it is important:	What carers can do to support the person:
Some medicines can interact with clozapine	- Some medicines can interfere with clozapine and make clozapine levels go up or down - This could give them more side effects, or stop their clozapine working properly - Some medicines have side effects that add to the clozapine's side effects - Illicit substances such as 'ice' and 'pot' can make it harder for clozapine to work.	- If the person starts any new medicines, make sure that the doctor, dentist or pharmacist knows all the other medicines they take, including clozapine - It is useful to know that side effects of different medicines can add to each other e.g. be careful with other medicines that have side effects such as constipation, dry mouth, finding it hard to pass urine, confusion, a fast heart beat and blurred vision. These can become a real problem - Also be careful with antibiotics.
Smoking and caffeine affect clozapine levels in the blood	- Cigarettes and tobacco can reduce clozapine levels (it is the hydrocarbons in the smoke that interact with the clozapine, rather than the nicotine) - Caffeine (in coffee, tea, Coca-Cola®, energy drinks, and chocolate) can increase clozapine levels - If there is any change in how much the person smokes or how much caffeine they take, their mental health team need to know (ideally before they make the change) - This is because the change might lead to under-treatment or side effects.	- Carers can support the person by keeping an eye on whether their smoking habit or caffeine intake changes - Quitting smoking is a big possible problem - You can remind them to let their mental health team know of any changes so they can review their clozapine dose - Stopping smoking will have big health benefits, but it is vital to monitor clozapine and symptoms during this time - The interaction is with smoking and tobacco, not with nicotine replacement therapy (NRT), so even if someone uses NRT you need to tell the health team. You can also see our factsheet on clozapine and smoking, which explains why smoking changes can be dangerous, and what to do about it.
Clozapine causes some other important side effects that carers should be aware of: <i>(This is not a full list of side effects; for more information see our other clozapine leaflets and fact sheets, or the manufacturer's leaflet)</i>		
Weight gain and increased risk of diabetes	- It is very likely that the person will put on weight when taking clozapine - If they can prevent this weight gain, especially early in treatment, this will reduce the risk of diabetes and heart problems - Support from you encouraging living a healthier lifestyle can be very useful.	- A healthier lifestyle can include a good diet including plenty of fruit and vegetables, plus regular exercise - You can support the person to go to their GP practice at least once a year for checks on their blood pressure, weight, blood glucose and blood fats - And help make sure any abnormal results are acted upon - Consider getting a referral from your GP for Medicare subsidised appointments with a dietician and exercise physiologist (try and find ones that have some experience in mental health). You can also see our factsheet on weight gain, and metabolic syndrome (which includes diabetes).

Page 3/4

<https://www.choiceandmedication.org/wadoh/>

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.



What the issue is:	Why it is important:	What carers can do to support the person:
Constipation or a slow gut	• Clozapine slows down the passage of food through the gut • This can cause constipation and sometimes a blockage in the bowels • This can be very serious. It can actually even be fatal But most people with constipation will not actually mention it, even when asked - sometimes a blockage can occur with no prior reports of constipation • Many people who take clozapine need regular laxatives to help keep their bowels moving smoothly.	• Ask people taking clozapine every few days about their bowel movements, and act on any reports of constipation • A good rule of thumb is to ask if they have done a poo in the last two days (a proper evacuation) • Carers should also be alert if the person complains about being sick, stomach pain, bloating, feeling uncomfortable, not feeling hungry or not eating • Contact the GP in the next day or so if you're worried, explaining that the person takes clozapine. See our fact sheet on clozapine and constipation. This will help you work out if anything is wrong and what to do about it. It can also, if needed, encourage healthcare professions to take constipation seriously.
Heart problems, especially in the first three months	• Thankfully, serious heart problems are rare with clozapine • It is, however, important to be aware of these in case they occur, especially during the first couple of months.	• You can help the person next day or so if they get short of breath (more than usual) when they try to do something, find it hard to sleep lying on their back, or are sweating, have heart palpitations, a fast heartbeat, 'flu-like symptoms or chest pain.
Feeling sleepy, sedated or tired	• This is an important side effect as it can stop the person trying to get on with their life • Sedation can also be a sign that the dose is too high, so talk with their mental health team for advice.	• It is very common for people to feel tired when treatment starts, but that it can get a bit better over time • It is sometimes helpful if the dose is adjusted e.g. how the total dose is spread throughout the day (e.g. taking more at night or in the evening, and less in the morning can sometimes help) • Carers can support the person not to drive or operate machines, and to avoid alcoholic drinks if they're sedated.
Neuroleptic Malignant Syndrome (often written as NMS)	• This is a very rare side effect, which usually only occurs when something else is going on e.g. they are very mentally unwell in hospital, they are physically unwell or take other medicines • It can be very serious so it is important that whoever is looking after them can spot that something's wrong.	Miss the next dose of clozapine and get medical advice in the next few hours if the person gets: • A high temperature (above 38.5°C) • Stiff muscles • Odd or irregular heartbeats. They can take paracetamol. You can also see our factsheet on clozapine and NMS. This will help you work out if anything is wrong and what to do about it.

The small print: This leaflet is to help carers understand more about how to help people stay well on clozapine. You must also read the manufacturer's Consumer Medicine Information (CMI) Leaflet. You may find more on the internet but beware as internet-based information is not always accurate. Do not share medicines with anyone else.

V07.03 [PM/RW/SRB 6.2020] ©2020 Mistura™ Enterprise Ltd (www.choiceandmedication.org). Choice and Medication™ indemnity applies only to licensed subscribing organisations and the personal use by that organisation's service users and carers. Use by non-subscribing organisations is prohibited.

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix N: Fact sheet stopping smoking



Government of Western Australia
Department of Health

Handy Fact Sheet Stopping smoking if you are taking clozapine

The problem:	Why it is important:	How you can help yourself:
- If you stop smoking, or cut down, you may get many more side effects from clozapine - This can be very serious	A change in smoking can make you very unwell	- Do not stop smoking suddenly - If you want to stop smoking, you must get help from your health professional - Your clozapine dose will need to be lower

What is the problem with smoking and clozapine?

- Smoking (including smoking cannabis) reduces the amount of clozapine in your body
- If you stop smoking, the amount of clozapine in your body can go up**, sometimes by as much as double (even higher if you also take some other medicines)
- You may get more side effects - this can be very serious.

Is smoking on clozapine important?

- Yes.** You may become very unwell if you stop or change your smoking
- Many people have become toxic on clozapine when they stopped smoking
- A few have even died
- Your doctor, pharmacist or case manager can help you stop smoking safely.

What causes this problem?

- Clozapine is broken down (metabolised) in the body by enzymes in the liver
- One of these enzymes is called CYP1A2 (or sometimes P450-1A2)
- Chemicals in the smoke get into the body and make this enzyme work faster
- That means the enzyme breaks clozapine down quicker, so you get less effect from it.

If you have the same dose of clozapine and your daily smoking habit doesn't change, there is no big problem. Other than smoking not being very healthy of course. The problem comes if you stop or start smoking.

When does the problem occur?

Stopping smoking:

- If you stop smoking your blood level of clozapine will go up**
- The effect can be huge, sometimes the same as almost doubling your dose
- This effect can be even more marked if you are also taking a medicine called sodium valproate
- You will then get more side effects from clozapine
- This can happen over 3-5 days and gets even worse over 2-4 weeks.

If you switch to any of these it will have the same effect as stopping smoking:

- E-cigarettes
- Nicotine patches, gum, sprays and lozenges
- Chewing tobacco and snuff.

Starting smoking:

- If you start smoking your blood levels will go down**
- This is because when you smoke, chemicals in the smoke get into the lungs and then your body and make the CYP1A2 enzyme work faster
- If you start smoking your levels of clozapine can start to fall in 3-5 days or maybe even sooner.
- Clozapine may not work as well
- You may become unwell again and your symptoms may come back.

<https://www.choiceandmedication.org/wadoh/>

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.



How will I know if my clozapine levels go up?

If your clozapine levels go up you may:

- Feel more sleepy
- Get more constipated or bugged up (see your doctor if you haven't done a poo for more than 3-4 days)
- Get more saliva in your mouth
- Feel very tired
- Feel dizzy when you stand up
- Have a seizure or fit.

If your levels go too high
you might also have a seizure or fit.
If this happens you **must**
see your doctor straight away.

What should I do if I want to stop smoking?

- Make sure your case manager, pharmacist or doctor knows
- Before you stop have a blood test to check on the amount of clozapine in your blood
 - This is a different test to your regular blood tests. The blood sample is usually sent away. The result may take a week or two to come back
- Stop smoking slowly over several weeks if you can
- If you are cutting down first, there may not be too much of an effect until you get down to about 10 cigarettes a day. Cutting out the last 10 cigarettes a day seems to be the problem
- Beware! You may get more side effects
- Keep in contact with your case manager, or doctor. A dose change may well be needed.

Remember, if you switch to any of these it will have the same effect as stopping smoking:

- E-cigarettes
- Nicotine patches, gum, sprays and lozenges
- Chewing tobacco and snuff.

What should I do if start smoking again?

- Please don't!
- You **must** tell your doctor or case manager if you do start smoking again
- You may well need another change in dose.

How can I help myself?

- If you want to change your smoking you **must get advice** and support from your prescriber
- You will need to have a lower dose of clozapine - there are ways of working out what your new dose might need to be
- You will probably just need to reduce your dose slowly
- Have another blood test to see how much clozapine there is in your blood can help get the dose right for you.

If you see a smoking cessation advisor
you must say that you are taking clozapine.
Take this sheet with you

Lifeline provides 24hr telephone crisis support on 13 11 14 or visit www.lifeline.org.au for information & downloads

We also have a shorter version of this fact sheet

The small print: This leaflet is to help you understand more about how smoking affects clozapine. You may find lots more on the internet but beware as internet-based information is not always accurate. Do not share medicines with anyone else.

V10.01 L15 [SRB 11.2020] ©2020 Mistura™ Enterprise Ltd (www.choiceandmedication.org). Choice and Medication™ indemnity applies only to licensed subscribing organisations and the personal use by that organisation's service users and carers. Use by non-subscribing organisations is prohibited.

<https://www.choiceandmedication.org/wadoh/>

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.

Appendix O: Fact sheet clozapine and constipation



Government of Western Australia
Department of Health

Handy Fact Sheet Clozapine and constipation

What the problem is:	Why it is important to you:	How you can help yourself:
- Clozapine can cause bad constipation (bunged up) - Sometimes people don't take this seriously	- It can be very painful and sometimes fatal - It can nearly always be avoided - Not everyone realises how important constipation with clozapine can be	- Take a regular laxative if needed - Note down your bowel movements - If you get constipated make a fuss and get treatment

What is constipation?

- Constipation is when you want to poo (pass a stool) but can't, or find it hard to poo regularly
- Normally, you should probably go at least every two to three days, although people vary.

How will I know if I have constipation?

A normal stool (poo) should not be hard but should not be too soft. The symptoms of constipation are:

- A change from your usual pattern of how many times a week you go e.g. going less often
- A change in hardness of your stools e.g. to dry, hard, lumpy stools
- Straining to pass a stool
- Stomach aches that get better after a poo
- Feeling sick, less hungry, bloated or full
- Smelly wind.

See the Bristol Stool Chart below. Types 3, 4 and 5 are generally OK. Types 1 and 2 mean you are constipated.

Bristol Stool Chart - Developed at University of Bristol

Type.1		Separate hard lumps, like nuts (hard to pass)
Type.2		Sausage-shaped but lumpy
Type.3		Like a sausage with cracks on its surface
Type.4		Like a sausage, smooth and soft
Type.5		Soft blobs, clear cut edges (passed easily)
Type.6		Fluffy pieces, ragged edges, mushy stool
Type.7		Watery, no solid pieces. Entirely liquid

Lewis SJ, Heaton KW (1997). 'Stool form scale as a useful guide to intestinal transit time'. *Scand J Gastroenterol* 32 (9): 920-4.



Constipation can get so bad that it looks like "the runs" (diarrhoea) as stuff from further up the bowel 'leaks' past the blockage.

What causes slow gut and constipation?

Clozapine can cause 'slow gut' (called 'clozapine-induced gastrointestinal hypomotility'). It slows down the longways contractions of the bowel, so waste takes a lot longer to get pushed through. This can then cause a blockage, or the bowel can stop working properly. It is more common if you:

- Have just started clozapine (although life-threatening slow-gut can occur at any time)
- Take a high dose of clozapine or have high blood levels (e.g. due to stopping smoking)
- Are also having another medicine that causes constipation e.g. benztropine, hyoscine hydrobromide (Kwells®), codeine or iron
- Also have another illness e.g. the flu
- Get dehydrated, as your body takes more water in from your stools and they get harder.

Ask your doctor or pharmacist for more advice.

Why is constipation important?

You must take constipation seriously because it can lead to:

- Bowel problems e.g. the stools can get impacted (solid and stuck). If this happens you will need a 'manual evacuation' (done with a gloved finger and no fun for anyone) or an enema (liquid squirted up to help move the stools down)
- Paralytic ileus – this is where the bowel becomes paralysed. This can have serious effects. There are, sadly, many cases of people who have died from constipation due to clozapine. These were mostly where the symptoms were not spotted in time, either by a health profession **OR** the person themselves.

<https://www.choiceandmedication.org/wadoh/>

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	<i>Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure</i>			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.



What can I do to help myself?

- **Fibre:** Make sure you eat enough fibre, cereal or fruit. At least five portions of different fruit or vegetables ('roughage') a day is a good start. You can also eat oats, nuts, seeds, wholegrain cereals, bran and wholemeal foods. This can take a few weeks to work fully as the fibre needs time to build up the bulk, so please *only increase your roughage intake gradually*

Fruits such as apples, apricots, gooseberries, grapes, raisins, peaches, pears, plums, prunes, raspberries and strawberries all contain sorbitol. Sorbitol is a laxative.

- **Liquid:** Make sure you drink enough liquid. You should have **at least 2L** (about 3½ pints) a day. That's about 2-3L (3½ to 5 pints) for men and 2-2.5L (3½ to 4½ pints) for women. Plus a bit more if it's really hot

Do not ignore feelings of thirst. A good measure of how hydrated you are is the colour of your wee. It should be a light-yellow colour. It should not be clear or a darker orange. It's better to be closer to the lighter yellow.

- **Go when you can:** Do **not** ignore the urge to go, or it can make constipation worse
- **Movement:** Keep active and get some exercise e.g. walking, can help keep your gut moving too
- **Laxatives:** You must take laxatives every day if you are constipated, or prone to getting constipated. Think about taking a laxative every day *even if you are not constipated* because serious problems can occur in people who didn't even think they were constipated.
 - Take a macrogol or stool softener (e.g. Dulcosoft®, Dendylax®, Transipeg®) or lactulose (a sweet-tasting liquid). These can be prescribed or bought from a pharmacy
 - It may be best to take them regularly e.g. every day. Any laxative taken should be a lubricant or softener (which can take several days to work). Do **not** take stimulants such as senna or bisacodyl regularly.

- **Be aware:** Keep a note of your bowel movements to make sure you know if there has been a change recently
 - Get help from a doctor before it gets too bad
- Write down which days you do a poo, the times, amounts and a Bristol Stool Chart number for each bowel movement. This may seem a bit odd but it *really* does help health professionals care for you
- Make sure family members/carers are aware of this serious side effect, so they will know if you need help.

When should I get help?

You should get help in the next few days if:

- Your stools are Type 1 or Type 2 on the Bristol Stool Chart
- Little bits of poo leak out between bowel movements
- You have not passed any stools for more than 3-4 days or if this happens regularly.

You should get urgent help if you have moderate to severe stomach pain or discomfort that lasts over an hour, especially if you also have:

- Swollen or distended stomach
- Diarrhoea/the runs (especially if there is blood in your stools or poo)
- Sickness or vomiting (especially if it smells of poo)
- Any signs of an infection e.g. a temperature
- A fast heart beat
- Malaise (feeling tired, lazy, dizzy or generally unwell).

If you have any of these symptoms, you must see your doctor or even go to hospital. Show them this leaflet. Remember, people have become very unwell from slow gut even when they didn't seem to be constipated.

☎ **Lifeline** provides 24hr telephone crisis support on 13 11 14 or visit www.lifeline.org.au for information & downloads

We also have a shorter version of this fact sheet

The small print: This leaflet is to help you understand more about constipation from clozapine. You may find lots more on the internet but beware as internet-based information is not always accurate. Do not share medicines with anyone else.

V08.05 L14 (SRB 11.2020) ©2020 Mistura™ Enterprise Ltd (www.choiceandmedication.org). Choice and Medication™ indemnity applies only to licensed subscribing organisations and the personal use by that organisation's service users and carers. Use by non-subscribing organisations is prohibited <https://www.choiceandmedication.org/wadon>

RESPONSIBILITY				NATIONAL STANDARDS
First Issued	Mar 2019	Revisions	NA	Mental Health National Standards 1.2, 1.3, 1.10, 1.12, 2.4, 6.7, 6.8, 6.9, 6.13, 8.4, 10.4.1, 10.5.7, 10.5.8.
This version	Dec 2022	Revision Due	Dec 2025	
Responsibility of	Co-Directors Mental Health	Endorsed by	Drugs & Therapeutics Committee	
Title	Clozapine Therapy: Initiation, Continuation and Ceasing (Mental Health) Policy and Procedure			OCP Standard Clinical Care Assessment, patient and carer involvement, physical health of mental health patient.